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Water Spray Fire Suppression Tests Comparing Gasoline-Fuelled and Battery Electric Vehicles

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Abstract. The increased use of electric vehicles has raised a concern about the performance efficiency of water spray fire suppression systems (often denoted "drencher systems") typically installed on ro–ro cargo and ro–ro passenger ships. A test series was conducted involving testing of two pairs of geometrically similar gasoline-fuelled and battery electric vehicles in test conditions as equivalent as possible. During testing, key parameters such as the heat release rate, the gas temperature above the vehicle and the surface temperature of target steel sheet screens at the sides of the vehicle were measured. Fire ignition was arranged in such a way that the gasoline fuel or the battery pack was involved at the initial stage of the fire. It is concluded that fires in the two types of vehicles are different but have similarities. However, a fire in a battery electric vehicle does not seem to be more challenging than a fire in a gasoline-fuelled vehicle for a drencher system designed in accordance with current international recommendations.

Keywords: Internal combustion engine vehicles, Battery electric vehicles, Sprinkler systems, Fire suppression, Ro–ro spaces, Ships

1. Introduction

Electric vehicle (EV) deployment has been growing rapidly over the past 10 years and will likely continue to grow. Basically, all car manufacturers in the world are introducing new fully electric or hybrid vehicles, partially in response to increasing efficiency and emissions standards. Large-scale free-burn fire tests [1–3] indicate that the peak heat release of a battery electric vehicle (BEV) fire is comparable to that of a vehicle with an internal combustion engine (ICEV), given similar sized vehicles. From these tests it is noted that the involvement and the time to involvement of the fuel tank and the battery pack of the vehicles have an influence on the fire growth rate and the severity of the fire. Kang et al. [4] conducted a series of large-scale fire tests focused on the understanding of thermal behaviours of battery electric vehicle fires. It is concluded that the characteristics of BEV fires are comparable with those of conventional passenger cars and that the combustibles

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in the battery pack present a minor contribution to the overall magnitude of BEV fires.

However, there is limited information available about the fire suppression performance of sprinkler systems. Water spray systems (often denoted ''drencher systems'') are typically installed on ro–ro cargo and ro–ro passenger ships. Roll-on/roll-off (ro–ro) ships are designed to carry wheeled cargo such as passenger cars, freight trucks, buses, and motorcycles, which are driven on and off the ship on their own wheels. The ships are large, with ro–ro spaces that extend the full length and width of the ship. The drencher system consists of deluge sections with open water spray nozzles at the ceiling. A deluge section (along the lanes) should not be less than 20 m in length, usually covers the full width of the space, and the system should be designed for the simultaneous activation of the two adjacent deluge sections with the greatest hydraulic demand. The minimum required water discharge density for spaces having a free height in excess of 2.5 m but equal to or less than 6.5 m is 10 mm/min. The sections are usually manually operated in case of fire, although automatic systems are permitted. Concerns have arisen about whether these systems are able to control a fire in an electric vehicle and if the design of the system in terms of water flow rates needs to be increased.

The objective of the tests was to compare the fire suppression performance of a deluge water spray system for fires involving ICEVs and BEVs in test conditions as equivalent as possible. The tests simulated a ro–ro space having a ceiling height of about 5 m and the system design in terms of water discharge densities corresponds to the design recommendations in MSC.1/Circ.1430/Rev.2 [5] given above.

2. The Vehicles Used in the Tests

Two pairs of vehicles, i.e., a total of four vehicles, were used in the tests. Each pair of vehicles was chosen to be as similar as possible, except for the powertrain, refer to Table 1. All vehicles were sport utility vehicles (SUVs). The make and model of the vehicles are not provided, but all vehicles are considered representative of modern vehicles in the marketplace.

ICEV1 and BEV1 were of the same make and similar except for the powertrain. BEV1 was geometrically slightly smaller but was about 30% heavier than the ICEV1, mainly due to the weight of the battery pack. It is also noted that BEV1 was built on a dedicated BEV platform with a simple flat rectangular battery pack design, with several battery modules positioned side by side between the wheel axles. The vehicle had Lithium Nickel Manganese Cobalt oxides (Li-NMC) pouch cells. BEV1 had a huge glass panel that stretches almost over the entire width of the roof.

ICEV2 and BEV2 were of the same make and were basically identical except for the powertrain as the vehicles are designed using a modular platform that allows different powertrains. The electric version is about 20% heavier, mainly due to the weight of the battery pack. The vehicle had Li-NMC prismatic cells, with the battery modules primarily arranged in one pack beneath the area of the back seat and in one pack beneath the area of the front seats.

3. The Positioning of the Vehicles

Six steel sheet trays were positioned at the floor of the fire test hall, symmetrically under the four water spray nozzles described below. Each tray measured 2500 mm (W) by 1000 mm (L) by 150 mm (H). The long sides of the trays were joined using bolts, long side-to-long side, to provide a total area of 2500 mm (W) by 6000 mm (L), equalling 15 m^2. The outer rim height of the trays was 150 mm, but the inner tray rims were designed with a lower height of 75 mm at the long sides that faced each other.

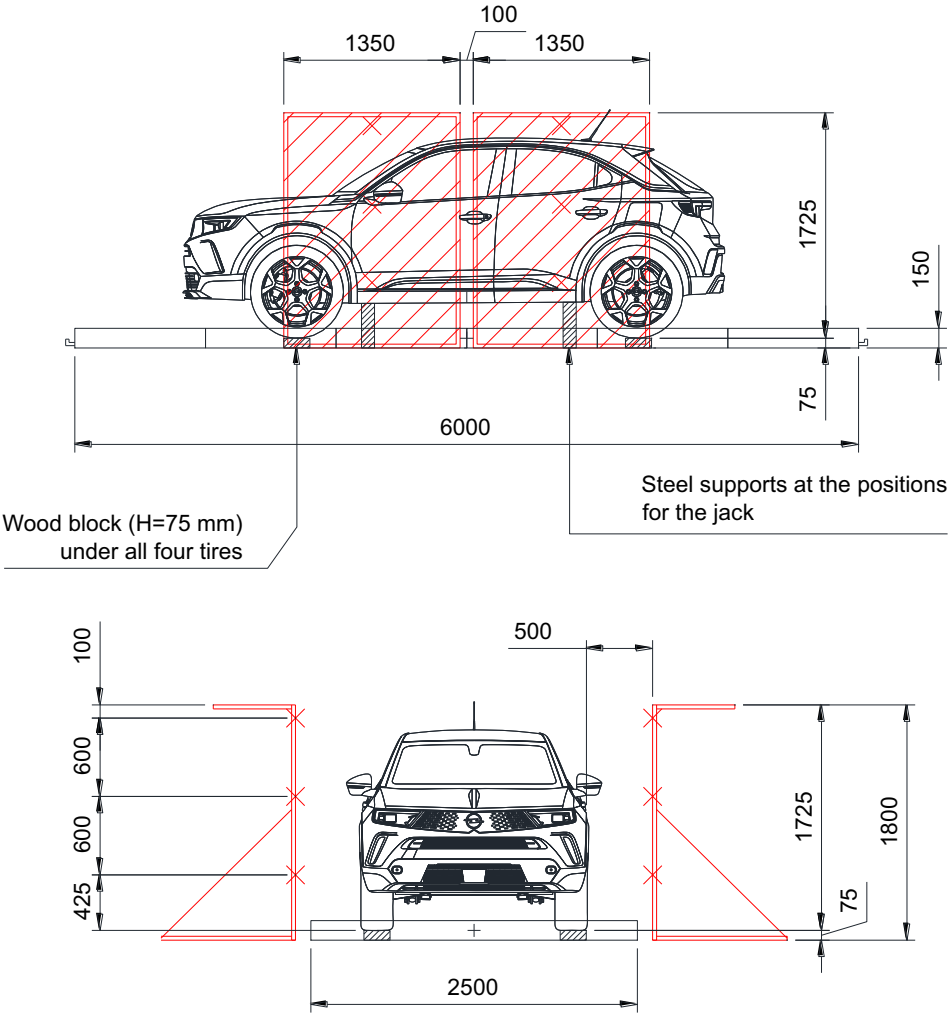


Figure 1. The arrangement of the vehicle in the tray filled with water, with the steel sheet screens to the sides. All measures in mm.



Figure 2. The spray pattern of the four water spray nozzles used in the tests.

Prior the tests, the trays were filled with 75 mm of potable water to achieve an unbroken water surface. Four goose-necks at the short sides ensured a constant level of water despite water being discharged into the trays by the over-head water spray nozzles. The trays were used to contain a spill of gasoline fuel and/or engine oil. The layer of water contributed to a completely smooth and horizontal surface over which the liquid fuel could uniformly spread out.

The vehicle under test was positioned symmetrically in the tray. Each of the tires was positioned on a small wood block that was 75 mm in height, making the vertical distance from underside of the vehicle to the water surface equal to the actual distance measured to a road surface. In addition, supports were placed at the jack brackets to prevent the vehicle from tilting or collapsing during a test. The intention was to improve the test-to-test repeatability by improving the stability of the vehicles. Figure 1 shows the arrangement.

4. The Preparation of the Vehicles

Vehicle components with the potential to explode when heated and become projectiles were either removed or punctured. All airbags were removed, and shock absorbers and hydraulic dampers were punctured. The air pressure in all tires was



Figure 3. ICEV1: The fire size at 01:12 (min:s), at the start of the application of water from the overhead water spray nozzles, viewed from a position in front of (left) and behind (right) the vehicle.



Figure 4. ICEV1: The fire size at the most intense stage at about 03:30 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.

released. The 12 V battery of all vehicles was removed to avoid any unnecessary contamination by lead in the run-off water.

The fuel tank was filled with unleaded 95-E10 to 90% of the nominal tank capacity. This gasoline quality is a fuel which contains up to 10% of ethanol, in accordance with the latest (from 2021) European specifications. A small (15 mm)



Figure 5. BEV1: The fire size a few seconds prior the start of the application of water from the overhead water spray nozzles at 12:40 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.



Figure 6. BEV1: The fire size at the most intense stage at about 19:00 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.



Figure 7. ICEV2: The fire size shortly after the start of the application of water from the overhead water spray nozzles at 00:58 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.



Figure 8. ICEV2: The fire size at the most intense stage at about 04:00 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.



Figure 9. BEV2: The fire size a few seconds prior and after the start of the application of water from the overhead water spray nozzles at 16:45 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.

circular hole was drilled at the bottom of the plastic fuel tank, close to the longitudinal centreline of the vehicle. The hole was fitted with a rubber plug, which was disconnected at the start of the test to allow an outflow of fuel over the water surface of the tray. The outflow was immediately ignited with a torch.

The propulsion battery pack was charged to 90% of its useable capacity. A small circular (22 mm diameter for BEV1) and square (30 mm × 30 mm for BEV2) hole was made through the protective plate under the battery pack. This allowed the penetration of one of the battery modules at the rear, left-hand side of the vehicle by a nail. The nail was pointed, had a diameter of 20 mm and was driven by a pneumatic cylinder placed inside the tray underneath the vehicle. The penetration depth was about 70 mm. If the combustion gases did not self-ignite, electric igniters positioned close to the nail and marshals that were placed at the top edge of the rim of the fire tray were supposed to ignite the gases.

All doors and windows of the vehicle were closed at the start of the test.

5. The Deluge Water Spray System

Four water spray nozzles were installed in a hydraulically balanced pipe-work, having a nozzle spacing of 3.05 m by 3.05 m. Each of the nozzles covered an area of 9.3 m^2. The pipe-work was constructed from DN50 (2″) steel pipe. A pressure transducer was installed at the end of one of the branch lines. The distribution line of the pipe-work had a solenoid valve that was remotely operated. The flow



Figure 10. BEV2: The fire size at the most intense stage at about 24:40 (min:s), viewed from a position in front of (left) and behind (right) the vehicle.

Table 1
The Vehicles Used in the Tests

	ICEV1	BEV1	ICEV2	BEV2
Model year	2022	2022	2021	2021
Type of vehicle	Compact SUV	Compact SUV	Subcompact crossover SUV	Subcompact crossover SUV
Curb weight (kg)	1614	2120	1295	1598
Drive	Front wheel drive	Rear wheel drive	Front wheel drive	Front wheel drive
Transmission	Manual (6-speed)	Automatic (single-speed)	Manual (6-speed)	Automatic (single-speed)
Fuel or type of battery	Gasoline	Lithium-Ion pouch cells	Gasoline	Lithium-ion prismatic cells
Battery cell chemistry	Not applicable	Li-NMC	Not applicable	Li-NMC
Fuel tank or battery capacity	58 l	82 kWh (total) 77 kWh (usable)	44 l	50 kWh (total) 45 kWh (usable)
Amount of fuel and charge level used in the test	52.2 l (90%)	69.3 kWh (90%)	39.6 l (90%)	40.5 kWh (90%)

rate of water was controlled via a diaphragm control valve connected to a compressed air supply. The control valve provided a constant pressure downstream of the valve, irrespective of the inlet pressure. However, the desired water flow rate

Table 2
Heat Release Rate Results, as Recorded Prior to or After the Start of the Application of Water

	ICEV1	BEV1	ICEV2	BEV2
Date of test	Sept. 23, 2022	Sept. 21, 2022	Sept. 27, 2022	Sept. 29, 2022
Time to discharge of water (min:s)	01:12 (72 s)	12:40 (760 s)	00:58 (58 s)	16:45 (1005 s)
Time to end of discharge of water (min:s)	31:12 (172 s)	42:40 (2560 s)	30:57 (1857s)	46:45 (2805 s)
Peak total heat release rate (kW)	7978	2944	5324	1975
Peak convective heat release rate (kW)	3594	1195	2323	1138
Maximum 5-minute average convective heat release rate (MJ)	1467	633	1006	639
Total heat release (MJ) from fire ignition to end of water discharge	2637	2189	1784	1370
Total heat release (MJ) from fire ignition to the end of test	5241	4510	4765	4474

Table 3
Temperature and Heat Flux Measurement Results as Recorded Prior to or After the Start of the Application of Water

	ICEV1	BEV1	ICEV2	BEV2
Date of test	Sept. 21, 2022	Sept. 23, 2022	Sept. 27, 2022	Sept. 29, 2022
Mean water flow rate with the Standard Deviation (l/min)	372 ± 2.8	373 ± 2.7	371 ± 2.8	374 ± 2.7
Peak gas temp. above the vehicle (°C)	340	56	135	166
Maximum average surface temp. on right-hand side steel sheet screen (°C)	407	149	277	165
Maximum average surface temp. on left-hand side steel sheet screen (°C)	339	139	251	163
Peak heat flux to the right of the vehicle (kW/m^2)	98	7	44	5
Peak heat flux to the left of the vehicle (kW/m^2)	138	6	59	6
Peak surface temp. on Plate Thermometer in front of the vehicle (°C)	70	43	33	36
Peak surface temp. on Plate Thermometer behind the vehicle (°C)	88	87	78	76

of 372 L/min to obtain a 10 mm/min discharge density was fine-tuned by an operator if required. The nominal water pressure at each nozzle was 1.3 bar. Potable water was supplied from the public main without the need for a pump.

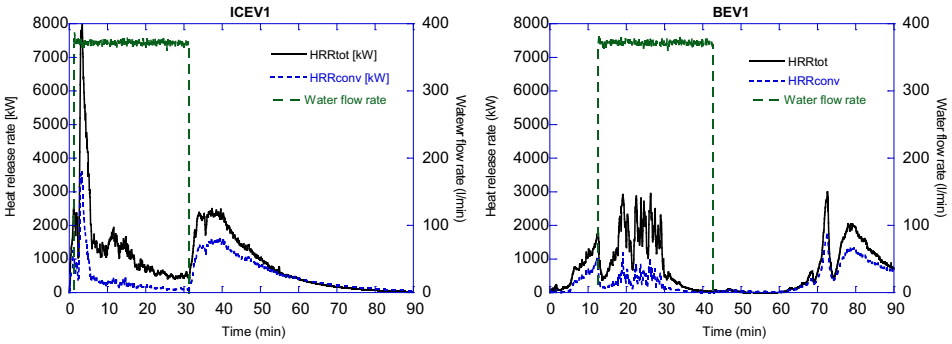


Figure 11. The total and convective heat release rate histories for ICEV1 and BEV1.

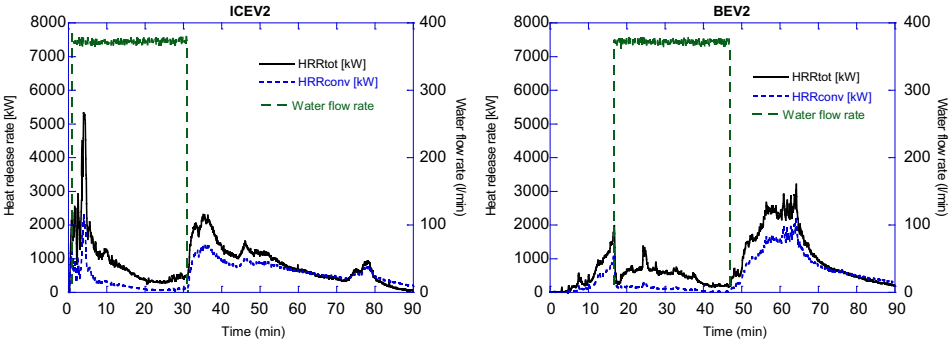


Figure 12. The total and convective heat release rate histories for ICEV2 and BEV2.

The nozzles used in the tests were open (non-automatic), pendent directional discharge water spray nozzles for fire protection purposes with a nominal K-factor of 80.6 (L/min)/$\sqrt{}$bar). The nozzles had an external deflector that discharged a uniformly filled cone of medium-velocity water droplets at a 180° spray angle. The nozzles used in the tests had no nozzle strainer.

The nozzles were installed with their frame arms parallel with the system branch lines, and thereby perpendicular to the long sides of the vehicle. The nozzles were positioned approximately 5.0 m above the water surface in the tray, as measured to the deflectors. This reflects a similar clear height in a ro–ro space on a ship. Figure 2 shows the spray pattern of the four nozzles.

6. Fire Test Procedures

The intent was to initiate fire in such a way that the liquid fuel or the battery pack was involved at the initial stage of the fire. This was done by making a mechanical penetration of the fuel tank and the battery, combined with small fire

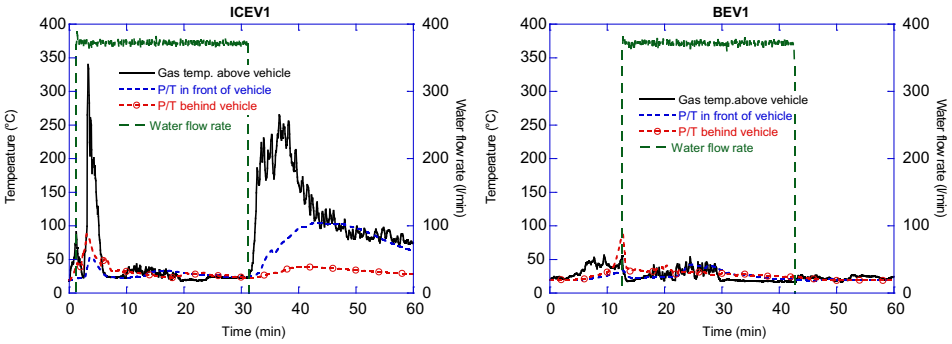


Figure 13. The gas temperature above the vehicle and the surface temperatures of the Plate Thermometers (P/T) in front of and behind the vehicle for ICEV1 and BEV1.

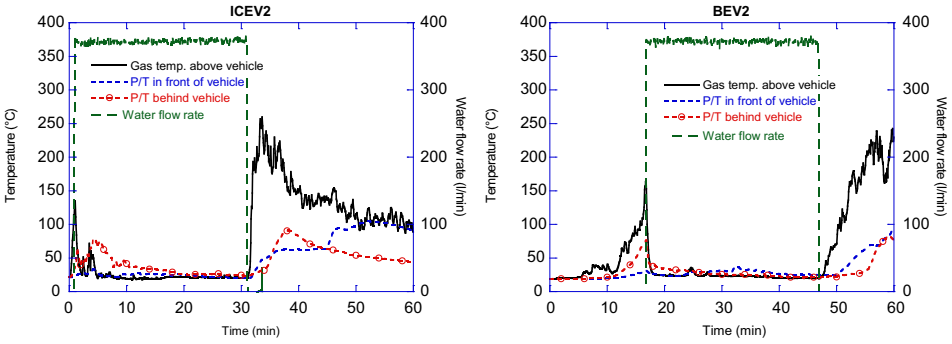


Figure 14. The gas temperature above the vehicle and the surface temperatures of the Plate Thermometers (P/T) in front of and behind the vehicle for ICEV2 and BEV2.

ignition sources that ensured that fire ignition occurred. Although it can be argued that this type of fire ignition scenario is extremely rare, the approach was required to obtain a straightforward comparison of the two distinct types of vehicles.

The application of water was manually initiated at a convective heat release rate of 1 MW, which corresponded to a total heat release rate of about 1.5 MW. This threshold was chosen to ensure that the fire had indeed involved the vehicle and at a time when continued fire growth was to be expected.

Water was discharged for 30 min. Thereafter, the post-application fire re-growth was documented by allowing the vehicle to be completely consumed in the fire. The approach facilitated the handling (scrapping) of the vehicles after the tests and provided an indication of the degree to which the fire was controlled by the application of water.

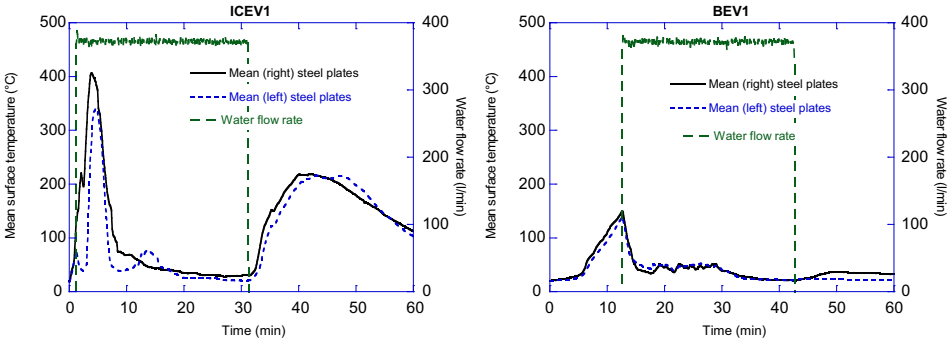


Figure 15. The mean surface temperature on the steel sheet screens to the right and left of the vehicle for ICEV1 and BEV1.

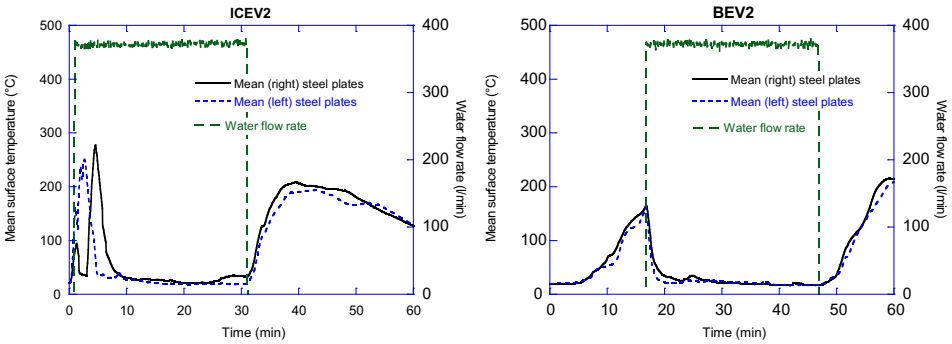


Figure 16. The mean surface temperature on the steel sheet screens to the right and left of the vehicle for ICEV2 and BEV2.

7. Instrumentation and Measurements

The following measurements were undertaken during each test:

7.1. Heat Release Rate Measurements

The tests were conducted under a large-scale calorimeter having a hood connected to an evacuation system capable of collecting all the combustion gases produced by the fire. The hood is 6 m in diameter with its lower rim 8 m above the floor. The calorimeter is described in [6]. Gas temperature, velocity, the generation of gaseous species such as CO_2 and CO, and the depletion of O_2 were measured in the exhaust duct. All measurement equipment in the exhaust duct and the calculation model for the heat release rate were based on the requirements in ISO 9705-1:2017 [7].

7.2. The Gas Temperature Inside the Vehicle

The gas temperature (C21) was measured inside the vehicle with a sheathed (Ø = 1 mm) thermocouple. The thermocouple was positioned between the head-rests of the front seats.

7.3. The Gas Temperature Above the Vehicle

The gas temperature (C22) above the vehicle was measured with a sheathed (Ø = 1 mm) thermocouple. The thermocouple was positioned 100 mm below the system pipe-work, at the connection point between the distribution pipe and the branch lines, i.e., above the mid-point of the vehicle. The vertical distance from the water surface in the tray to the thermocouple was approximately 5 m.

7.4. The Surface Temperature on Steel Sheet Screens to the Sides of the Vehicle

Two steel sheet screens were positioned at each long side of the tested vehicle. Each screen was sized 1350 mm (L) by 1800 mm (H) and had a nominal thickness of 1 mm. The screens were positioned symmetrically with respect to the wheel axles and 100 mm apart. The horizontal distance to the side of the vehicle was 500 mm. Each steel screen had a horizontal 600 mm wide overhang that prevented water from wetting its back side.

The surface temperatures at selected locations of the steel sheet screens were measured using wire thermocouples (Ø = 0.5 mm) that were spot-welded to the back side of the screens (C23–C34). Each screen had a column of three thermo-couples positioned along its vertical centreline. The vertical distance from the top of the screen to each thermocouple was 100 mm, 700 mm, and 1300 mm, respectively. The bottommost thermocouple was thereby 425 mm above the water surface of the tray.

The surface temperature measurements reflect the temperature of an uninsulated body panel of a vehicle, and the measurement points were positioned to capture exposure by flames from both the underneath of the vehicle (as fire in the gasoline fuel spill or battery pack) as well as from flames in rubber tires and through broken side windows.

7.5. Heat Radiation Measurements

The heat radiation was measured with heat flux meters facing the long side of the vehicle. The devices were positioned 1125 mm above the water surface in the tray and at a horizontal distance of 500 mm from the right (C35) and left (C36) side of the vehicle. The position was between the steel sheet screens described above, at the 100 mm gap. The height of the devices above the floor corresponded to the approximate midline of the side windows of the vehicle.

7.6. Plate Thermometer Measurements

One Plate Thermometer was positioned along the longitudinal centreline of the vehicle, at a horizontal distance of 1500 mm from the front (C37) and rear (C38) of the vehicle, respectively. The vertical distance measured to the centre point of the water surface in the tray was 750 mm.

The Plate Thermometer consists of a 0.7 mm thick Inconel 600 steel plate with a front face measuring 100 mm by 100 mm. A sheathed thermocouple is spot-welded to the plate that is insulated on the backside. The device is sensitive to heat convection, but compared to a conventional thermocouple, significantly more sensitive to heat radiation [8, 9]. For these tests, the devices were primarily exposed to heat radiation as they were not directly touched by flames from the fire.

7.7. Measurements of the System Operating Pressure and Water Flow Rate

The system operating pressure (C39) was measured using a pressure transducer positioned at one of the system branches. The pressure transducer was positioned at the end of the pipe, i.e., there was a minimal static pressure difference between the nozzles and the transducer. The water flow (C40) rate was measured using a flow meter installed after the flow control device.

8. Fire Test observations

The following observations were made during the tests:

8.1. ICEV1

The initial fire development was fast as the gasoline fuel spread across the water surface. The application of water that started at 01:12 (min:s) temporarily limited the fire growth rate, refer to Figure 3, but at about 02:30 (min:s) the fire re-developed rapidly as the spill area increased in size. This is likely due to a larger burn-through of the plastic fuel tank. A peak total heat release rate of almost 8 MW was recorded at about 03:30 (min:s), refer to Figure 4. At that time, the fuel spill area extended almost to the front end of the tray. Visually, about 90% of the 15 m^2 surface area of the tray was burning. This stage was followed by a gradual decline of the fire size as the gasoline fuel was steadily consumed. At about 06:00 (min:s) the gasoline fuel was completely consumed, and the fire was located at the rear of the vehicle, including the rear tires. About 30 s earlier, flames were also observed through the front side window at the left-hand side (driver's side) of the vehicle, indicating that the interior of the passenger compartment was burning.

Following the burn out of the fuel spill, the fire size never exceeded 2 MW and the fire size was gradually reduced until the application of water was terminated.

At the time the water flow was stopped, the fire visually involved the engine compartment and the passenger compartment, with observed burn-through of the windscreen and the front side window at the left-hand side. Once the water flow was stopped, the fire re-developed and burned at a level of around 2.2 MW for

about 7 min before it slowly decreased in size. The increase in fire size was visually associated with an initial increase of the fire inside the passenger compartment, with flames projecting through the windscreen and the side windows. It was also observed that the side windows at the right-hand side had partly burnt through. During this stage, the paint on the hood started to blacken and the paint of the roof ignited and burned. It is noticeable that the paint on the hood and on the roof of the vehicle was virtually undamaged because of the application of water. The gap in the windscreen increased in size which allowed the fire in the passenger compartment to increase. The fire reached its peak during the stage when all windows had been completely damaged. To some extent, the fire in the engine compartment contributed, as flames were observed in a gap that opened between the hood and fenders. These flames eventually involved the front tires, and it is clear that this sequence of events extended the post-application peak total heat release rate.

8.2. BEV1

Fire ignition was immediate when the nail penetrated the battery module, with flames projecting from the left-hand side of the vehicle and generation of visible smoke. At 00:20 (min:s), smoke plumes at both sides of the vehicle appeared, originating from the rear wheelhouses, followed by short duration jet flames at 01:00 (min:s). At about 02:00 (min:s), plenty of burning melted plastic pieces from the low-side panels were observed. At about 04:45 (min:s) it was observed that all four tires and the wheelhouse liners start to become involved in the fire. At about 05:30 (min:s), the smoke plumes originating from the area above the rear tires turned into durable jet flames, which gradually increased the fire involvement of the rear tires. At 07:00 (min:s) flames were observed at the areas of the water drainage hole in front of the windscreen at the left-hand side of the vehicle. These flames grew in size and extended above the roof at 09:50 (min:s). At about 10:00 (min:s) the fire started to involve the plastic parts at the rear (left) end of the vehicle, which resulted in an increase in the fire growth rate. The rear was fully involved in the fire at 12:10 (min:s). At 12:40 (min:s) the application of water was initiated, refer to Figure 5. This resulted in a prompt reduction of the total heat release rate. However, from the measurements it is noticed that the fire gradually re-developed starting at about 14:45 (min:s). Visibility was obscured by the water spray and smoke, but at 16:00 (min:s), durable jet flames extending from the area above the rear tires were observed. At about 18:00 (min:s) a more rapid fire re-growth is noticed and the second total heat release rate peak at 19:00 (min:s) is associated with the fire involvement of the battery pack, the rear tires, and the interior of the passenger compartment, with flames through the side windows. Figure 6 shows the fire size when it was the most intense. After this stage, the fire size declined but re-developed again at 21:45 (min:s), as the fire inside the passenger compartment increased. Visually, it seemed that the contribution to the fire from the rear of the battery pack was small. The fire re-growth may be due in part to the involvement of the mid- and front battery modules. At 25:00 (min:s), it is observed that area under the front hood and front tires are

severely involved in the fire. The progressing fire spread towards the front of the vehicle lasted for about 11 min, from about 18:00 (min:s) to about 29:00 (min:s), where the fire size fluctuated between about 1 MW and 3 MW. The fire size steadily decreased until the water flow was terminated at 42:40 (min:s). At that time, only small flames were visible at the underside of the vehicle, with smoke gradually increasing.

Once the water flow rate was terminated, the fire did not re-develop until about 61:00 (min:s), i.e., after about 18 min. It was observed that the glass sunroof remained intact due to the application of water, but the rear window and most of the windscreen had broken. Visually, the fire re-developed in an area close to the driver's seat. At about 68:00 (min:s), the glass sunroof above the driver's seat started to gradually break down and the fire size increased as it involved the unwetted interior. The fire was spreading towards the right-hand side of the vehicle and peaked at almost 3 MW at 73:00 (min:s), before it started to decrease. The fire spread inside the passenger compartment continued, which led to a second peak of about 2 MW at 79:00 (min:s), when the area of the luggage compartment was involved in the fire. After this stage, the fire size gradually declined as the combustibles burnt out.

8.3. ICEV2

The fire ignited immediately, and the initial fire development was fast as the gasoline fuel spread across the water surface, with higher flames initially observed at the left-hand side of the vehicle. The application of water started at 00:58 (min:s), refer to Figure 7. To some extent this temporarily reduced the spill fire size at the right-hand side of the vehicle. During this stage, the fire size varied between 2 and 3 MW. At about 03:00 (min:s), the fire size increased rapidly as the spill area increased in size, with higher flames appearing at the right-hand side again. This increase is likely due to a larger burn-through of the plastic fuel tank. A peak total heat release rate of approximately 5,3 MW was recorded at about 04:00 (min:s), refer to Figure 8. Visually, the spill fire was located at the rear of the fire tray and at the right-hand side of the vehicle. The spill area visually extended at most up to the front tires. The peak lasted for about 30 s, thereafter the fire gradually declined as the gasoline fuel was steadily consumed. At 05:10 (min:s) it seemed that most of the gasoline fuel had been consumed and the fire was located at the rear, including involvement of the rear tires. At this time, burn-through of the rear side window on the left-hand side was observed.

This intense stage was followed by a steadily decreasing fire size as the plastic parts at the rear of the vehicle and the rear tires were consumed. At 23:00 (min:s) the fire basically involved only the rear tires and the engine compartment, where the fire was completely shielded from the application of water. At about 24:00 (min:s), flames were observed from the front side window at the right-hand side of the vehicle. These flames gradually increased in size, likely as the damaged area of the window increased. At 26:00 (min:s), these flames intermittently extended between 0.5 m and 1 m above the roof of the vehicle and occasionally became so large that they touched the steel sheet screen to the right. At 31:10 (min:s),

flames were also observed from a damaged area at the front side window on the left-hand side. These flames were growing larger and when the application of water was stopped at 30:58 (min:s) it was observed that the fire size immediately increased inside the passenger compartment of the vehicle, with flames extending through the side windows and shortly thereafter through the windscreen that rapidly burnt through to its full extent. When visibility improved, it was also observed that the rear window had broken. The most intense stage of the fire inside the passenger compartment lasted for about 7 min and peaked at around 2.3 MW before it slowly declined.

At about 42:00 (min:s) the fire started to spread towards the front of the vehicle, with flames moving underneath the right-hand side of the hood and in the front right wheelhouse. At about 46:30 (min:s), almost all of the plastic parts of the front were involved in the fire, whilst part of the passenger compartment was still burning. This resulted in a short, second peak of about 1.5 MW and a slow decay in the fire size as the combustibles at the front and inside the passenger compartment were consumed. The trend of steady decay was broken at about 73:00 (min:s) when the fire in the front tires and inside the front part of the compartment intensified, leading to a peak of almost 900 kW at about 78:00 (min:s). Thereafter, all combustibles were progressively consumed.

8.4. BEV2

The first flare-up occurred at 00:57 (min:s) but no sustained burning was observed. At 02:12 (min:s), burning drops of melted plastic appeared at irregular intervals from the underside of the vehicle. These droplets fell into the layer of water, which prevented a pile of melted plastic to form. The number of droplets gradually increased with time and at about 03:00 (min:s) several droplets per second was observed. The melted plastic formed a burning pile at the top of the pneumatic jack that extended through the layer of water. At 03:47 (min:s), the flammable gases ignited, with flames going from the underside and up the right-hand side of the vehicle. Moments later, these flames disappeared. At 04:30, flames were observed again. These flames extended from the underside of the vehicle and stretched up the side of the back door and extended from the area above the rear tire at the left-hand side. Once again, the flames disappeared after a few seconds, followed by a flare-up underneath the vehicle. It was also observed that the amount of smoke increased. At 05:30 (min:s), burning plastic was flowing through the hole where the nail penetrated the battery pack. At this time visible smoke was also observed.

The major contribution to the fire after this stage came from burning parts of the undercarriage plastic cover panels that fell down from the underside of the vehicle. At 07:15 (min:s) a jet flame appeared from the left-hand underside of the vehicle and a few seconds later a similar jet flame appeared from a more central part of the battery pack. These jet flames were intense for about ten seconds before they vanished, but other (less intense flames) were observed shortly thereafter. Once these flames faded away the fire growth rate slowed down, with occasional jet flames being observed. The fire started to increase in size at about

11:00 (min:s), once it involved the plastic panels at the sides and started to spread to the front tires and plastic panels surrounding the front wheelhouses. At this stage, durable jet flames appeared from the rear wheelhouses, indicating a larger involvement of the rear battery pack. These jet flames involved the rear tires in the fire. At about 12:00 (min:s) flames were observed at the areas of drainage holes in front of the windscreen. These flames grew and the smoke observed through the grill indicates that the electric motor and the associated equipment became involved in the fire. The rear plastic panel fell down gradually at about 15:30 (min:s), increasing the size of the fire. The application of water started at 16:45 (min:s), refer to Figure 9. This resulted in a prompt reduction of the total heat release rate. Shortly after the application of water, the rear window burnt through, which may have exposed part of the interior to the water spray. But the opening also allowed air to enter, which increased the fire. At about 22:00 (min:s) the fires in the rear part of the battery pack and the rear tires were reduced, observed as less intense jet flames. At 24:00 (min:s), large flames at the front edge of the hood were observed, indicating fire spread that was shielded from the application of water. The fire was the most intense at about 24:40 (min:s), refer to Figure 10, when it visually involved the passenger and engine compartments. Flames from the rear wheelhouses do also suggest that the battery pack to some extent was involved in the fire.

Once the water flow was stopped at 46:45 (min:s), only small flames were observed at the underside of the vehicle in an area in front of and behind the rear left-hand tire, at the front wheelhouse to the left and inside the passenger compartment. It was observed that all windows had broken, which probably allowed access of water to the inside of the vehicle. Fire re-growth was observed shortly after the end of water application and initially involved fire spreading to unburnt combustibles (plastic) at the front of the vehicle. But the fire also progressed inside the vehicle and towards the rear. About a minute after the end of water application, short fire flare-ups were observed which may be due to the fire in combustible gases expelling from individual cells of the battery pack at the rear. These flare-ups continued at intervals between 30 s and 60 s as the fire inside the passenger compartment progressed towards the rear of the vehicle.

When all unburnt combustibles where involved the fire burnt intensely at a level of between 2 MW and 3 MW for around 8 min. At this stage, large flames were observed through all window openings, and at the front and the rear of the vehicle. During the final 5 min of this 8-min stage, intermittent flames were observed at the rear end and from the rear wheelhouses. These flames are probably associated with the burn-out of the rear battery pack. Although it cannot be fully confirmed, water ingress may have prevented the full involvement of the rear end battery pack during the stage water was applied. After this stage, starting at 64:30 (min:s), the fire declined as the remaining combustibles burnt out.

9. Test Results

Table 2 shows the heat release rate results, as recorded prior to or after the start of the application of water.

Figures 11 and 12 show the heat release rate histories for each pair of vehicles.

Table 3 shows the temperature and heat flux measurement results as recorded prior to or after the start of the application of water.

Figures 13 and 14 show the gas temperature above the vehicle and the surface temperatures of the Plate Thermometers (P/T) in front of and behind the vehicle. Note that the time scale extends to 60 min and not to 90 min as in the previous figures, to better capture the sequence of events when the water spray system was operating.

Figures 15 and 16 show the mean surface temperature of the steel sheet screens to the sides of the vehicle.

10. Discussion

10.1. The Fire Scenarios

The fire ignition scenarios are considered unlikely but were necessitated by the desire to initiate the fire either in the gasoline fuel or in the battery pack and thereafter allow the fire to spread to other combustible parts of the vehicle. Both fire ignition scenarios proved to work from the aspect that fire ignition was immediate in the flammable gases of the fuel spill fire as well as in the flammable gases escaping the battery pack, except for BEV2 where fire ignition occurred immediately but the presence of stable flames was a little delayed.

Thereafter, the fire scenarios of the two types of vehicles were quite different. The gasoline fuel spill fire in the ICEV tests developed rapidly and the application of water was initiated after about a minute. The fire in the battery pack of the BEVs developed slower, involved other combustibles gradually and the application of water was initiated after more than 12 min and 16 min, respectively.

The fires in the ICEVs were initially suppressed, but the flowing gasoline fuel spill fire caused more damage to the plastic fuel tank that resulted in an increase of the spill area and the peak total heat release rate. Plastic fuel tanks in passenger vehicles are designed to meet international fire test requirements [10] and should withstand a spill fire test scenario for 2 min, which correlates well with the observations of their integrity in these tests. The peak total heat release rate occurred during a period from between three and 4 min (ICEV1) and four and four and a half minutes (ICEV2). The peak total heat release rate of ICEV1 was higher than that of ICEV2, which correlates with the larger quantity of gasoline used in ICEV1. After the burn-out of the spill fire at about 6 min (ICEV1) and 5 min (ICEV2), fire continued in combustibles such as the tires, the plastic liners in the wheelhouses, undercarriage plastic cover panels, inside the passenger compartment and inside the engine compartment. These combustibles are completely or partly shielded from the application of water from the overhead water spray nozzles. During this stage of the continued 30-min application of water, the total

heat release rate gradually decreased as these combustibles were consumed. Once the application of water was terminated, fire re-developed inside the passenger compartment, partly as the extent of damage to windows increased. Additionally, the paint and external front parts of the vehicle that were not involved in the fire due to the application of water started to burn. The post-application peak total heat release rates were significant for both vehicles.

The fire in the BEVs had involved other combustibles to a larger extent when the application of water was initiated. Even though these combustibles were completely or partly shielded from the application of water, the fire was promptly suppressed. The fire re-growth experienced for BEV1 is primarily associated with the battery pack where the fire progressed from the rear part (where fire ignition took place) towards the front part. However, contributions from the passenger compartment were also observed. The progress of the fire from battery module to battery module is captured by regular peaks in the heat release measurement. For BEV2, the fire re-growth was not as significant and involved the passenger and engine compartments. Flames from the rear wheelhouses do also suggest that the battery pack was involved in the fire. Once the application of water was terminated, fire re-development was significantly slower in BEV1 (after about 18 min) than in BEV2 (basically immediately). The post-application fire re-growth involved unburnt combustibles such as the paint and front plastic panels as well as unburnt parts of the interior. The post-application peak total heat release rates were significant for both vehicles. For BEV1, it seems that the battery pack burnt out during the 30-min application of water. For BEV2, water entering through the window openings may have prevent the battery modules from being fully consumed. During the burn-out of the vehicle after the end of water application, it visually seems that a fire in the battery pack at the rear to some extent contributed to the overall fire severity.

It is concluded that the peak total heat release rate was significantly higher for ICEV1 than for BEV1, as well as for ICEV2 compared to BEV2, which is associated with the short, but intense, gasoline fuel spill fire. The maximum 5-min average convective heat release rate captures the severity of the fire over a longer period. The parameter value was the highest for ICEV1; more than twice as high as for BEV1 and BEV2. The value for ICEV2 was lower than that of ICEV1, as the vehicle was smaller with a smaller amount of liquid fuel.

The total heat release from fire ignition and to the end of water application was slightly higher for ICEV1 as compared to BEV1 and higher for ICEV2 as compared to BEV2. The same trends are observed for the entire test period, which included the burn-out of the vehicles.

10.2. The Performance of the Water Spray System

One of the most asked questions is that on the relative likelihood of vehicle-to-vehicle fire spread for battery electric vehicles versus traditional vehicles? These tests indicate that a large flammable liquid fuel spill fire could potentially involve adjacent vehicles; a large fuel spill burns very intensely, and the liquid could spread under adjacent vehicles. The mean surface temperatures of the steel sheet screens

to the sides of the vehicle were significantly higher during the ICEV tests than in the BEV tests due to the fuel spill fire. Once the fuel spill fire had burnt out, surface temperatures were reduced to moderate levels. The surface temperatures generated by the fire in the battery pack in the BEV tests were not noticeably higher than those in the ICEV tests during the stage when water was applied.

The degree of fire control was established by terminating the water flow and observing the time to fire re-growth and the magnitude of the fire. For both types of vehicles, the fire re-peaked at between 2 MW and 3 MW after the application of water was terminated. This is an indication that the total heat release rate of the fires was indeed reduced by the water spray, despite the fact that the fire was, to a large degree, shielded. Once the wetting and cooling of the body of a vehicle stopped, it was observed that damage to windows will occur or increase, paint will start to burn, and unburnt combustibles will ignite that contribute to fire re-growth.

The gas temperature above the vehicle was promptly reduced in the BEV tests to a level of 50°C or less during water discharge. For the ICEV tests, a short peak reaching 340°C (ICEV1) and 135°C (ICEV2) was observed after the application of water was initiated, which relates to the increase of the gasoline fuel spill fire. After the stage when the liquid fuel burns out, the gas temperature was reduced to less than 50°C.

None of the Plate Thermometer surface temperatures peaked at any significant levels in any of the tests. It is observed that the peak temperature of the device facing the rear of the vehicles was consistently higher than the device facing the front. This observation seems logical as the fire was started at the rear part of the vehicles. The heat fluxes measured during the tests with the BEVs were lower than in the tests with the ICEVs. But it should be recognized that the two devices were positioned at a height corresponding to the midline of the side windows, where the radiation from the fuel spill fires was high due to larger (higher) flames.

The performance of the water spray system would be influenced by the time to operate the system. The longer the time, the larger the fire. Although not specifically investigated, the performance results are likely also valid for cases when the fire is larger at the start of water application than in the tests. An increase in heat release rate for a single vehicle would require burn through of windows or glass sunroofs, which would expose the combustibles in the passenger compartment to the water spray and thereby improve system performance.

Manual fire-fighting efforts were not part of the study but based on the tests it can be argued that an intentional premature termination of a drencher system should be avoided. When the water spray system is turned-off, resources should be readily available to manage a re-developing fire.

11. Conclusion

Concerns about the performance of drencher systems in ro–ro cargo spaces on board ships have been raised regarding the increased number of battery electric vehicles being transported. The objective of these tests was to compare the fire

suppression performance of a deluge water spray system for fires involving ICEVs and BEVs in test conditions as equivalent as possible. The tests simulated a ro–ro space having a ceiling height of about 5 m with a system design in line with the recommendations in MSC.1/Circ.1430/Rev.2.

It is concluded that a fire in the two types of vehicles is different but have similarities. A fuel spill fire associated with an ICEV develops very rapidly, peaks high but burns out fast, whilst a fire starting in the battery pack of a BEV develops slower, is not as large but burns longer. The scenario of the fire in other combustibles, such as the tires, exterior and undercarriage plastic parts and inside the passenger compartment is similar.

The deluge water spray controlled the fire, irrespective of the type of vehicle, measured as a reduction of gas temperatures above the vehicle, the surface temperatures of adjacent steel sheet screens, the surface temperature of Plate Thermometers, and reduced heat radiation. Fire control was also documented by a reduction of the heat release during water application and clear fire re-growth with significant fire sizes once the application was terminated. The overall conclusion from the tests is that a fire in a BEV does not seem to be more challenging for the drencher system design given in MSC.1/Circ.1430/Rev.2 than a fire in an ICEV of comparable size. One concern is raised—a flammable liquid fuel spill fire could involve adjacent vehicles due to its intensity and as the liquid spill could spread under adjacent vehicles.

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Declarations

Conflict of interest All authors declare that they have no conflicts of interest.

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References

1. Lam C, MacNeil D, Kroeker R, Lougheed G, Lalime G (2016) Full-scale fire testing of electric and internal combustion engine vehicles, Fourth International Conference on Fire in Vehicle, Baltimore
2. Watanabe N, Sugaw O, Suwa T, Ogawa Y, Hiramatsu M, Tomonori H et al. (2012) Comparison of fire behaviours of an electric-battery-powered vehicle and gasoline-powered vehicle in a real-scale fire test, Second International Conference on Fires in Vehicles, Chicago
3. Willstrand O, Bisschop R, Blomqvist P, Temple A, Anderson J (2020) Toxic gases from fire in electric vehicles. RISE Rep 2020:90
4. Kang S, Kwon M, Choi JY, Choi S (2023) Full-scale fire testing of battery electric vehicles. Appl Energy 332:120497
5. MSC.1/Circ. 1430/Rev.2, Revised Guidelines for the Design and Approval of Fixed Water-Based Fire-Fighting Systems for Ro–Ro Spaces and Special Category Spaces, International Maritime Organization, 8 Dec 2020
6. Dahlberg M (1992) The SP industry calorimeter for rate of heat release measurements up to 10 MW, SP Report 1992:43, Swedish National Testing and Research Institute
7. ISO 9705-1:2017, Reaction to fire tests - Room corner test for wall and ceiling lining products—part 1: Test method for a small room configuration (ISO 9705-1:2016, IDT), Approved: 2017-02-15
8. Wickström U (1989) The plate thermometer—a simple instrument for reaching harmonized fire resistance tests, SP-REPORT 1989:03, Swedish National Testing and Research Institute, Borås
9. Wickström U (1994) The plate thermometer—a simple instrument for reaching harmonized fire resistance tests. Fire Technol 30(2):195
10. Addendum 33: Regulation No. 34, Revision 3, Uniform provisions concerning the approval of vehicles with regard to the prevention of fire risks, United Nations, 12 November 2015